

## XINING, China

WMO: 528660

Lat: 36.6644N

Lon: 101.7331E

Elev: 2410

StdP: 75.53

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -17.3      | -15.6 | -24.9                         | 0.5 | -6.0 | -23.3 | 0.6 | -6.8 | 5.6                   | -1.6 | 4.9 | -1.5 | 0.3                   | 20   | 0.519 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.8                   | 28.0            | 15.4 | 26.2 | 14.5 | 24.6 | 13.7 | 17.2                | 24.3 | 16.3 | 22.8 | 15.5 | 21.6 | 1.6                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 15.3                            | 14.6 | 19.5 | 14.4 | 13.8 | 18.8 | 13.5 | 13.0 | 18.1 | 59.2          | 24.6 | 56.0 | 22.6 | 53.1 | 21.8 | 20.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 5.4           | 4.2  | 3.3 | -20.4                      | 31.4 | 2.0                | 1.8 | -21.8                                              | 32.8 | -23.0      | 33.8 | -24.1      | 34.9 | -25.5      | 36.2 |     |     |
| (5)               |      |     | -20.8                      | 18.5 | 1.9                | 1.1 | -22.2                                              | 19.3 | -23.3      | 19.9 | -24.4      | 20.5 | -25.8      | 21.3 | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 7.0    | -6.5  | -2.6  | 2.9   | 8.9   | 12.7  | 15.8  | 18.0  | 17.4  | 13.2  | 7.4   | 0.8   | -5.0  | (6)  |  |
|                                                                        | DBStd       | 8.97   | 3.03  | 3.51  | 3.66  | 3.59  | 2.87  | 2.28  | 2.60  | 2.66  | 2.55  | 2.79  | 3.39  | 2.79  | (7)  |  |
|                                                                        | HDD10.0     | 1989   | 511   | 353   | 220   | 61    | 9     | 0     | 0     | 0     | 3     | 87    | 277   | 467   | (8)  |  |
|                                                                        | HDD18.3     | 4208   | 769   | 587   | 477   | 284   | 176   | 81    | 37    | 50    | 155   | 338   | 527   | 725   | (9)  |  |
|                                                                        | CDD10.0     | 877    | 0     | 0     | 1     | 26    | 92    | 174   | 248   | 230   | 99    | 7     | 0     | 0     | (10) |  |
|                                                                        | CDD18.3     | 54     | 0     | 0     | 0     | 0     | 0     | 4     | 27    | 22    | 1     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 660    | 0     | 0     | 1     | 9     | 30    | 88    | 312   | 203   | 17    | 1     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 127    | 0     | 0     | 0     | 1     | 2     | 8     | 75    | 40    | 1     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 1.1    | 0.9   | 1.2   | 1.5   | 1.6   | 1.4   | 1.2   | 1.0   | 1.0   | 0.9   | 1.0   | 0.9   | 0.8   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 424    | 2     | 2     | 10    | 23    | 55    | 68    | 83    | 86    | 66    | 23    | 4     | 1     | (15) |  |
|                                                                        | PrecMax     | 540    | 6     | 7     | 28    | 69    | 109   | 112   | 149   | 163   | 116   | 58    | 14    | 6     | (16) |  |
|                                                                        | PrecMin     | 275    | 0     | 0     | 1     | 5     | 22    | 38    | 45    | 38    | 14    | 8     | 0     | 0     | (17) |  |
|                                                                        | PrecStd     | 55     | 1     | 2     | 8     | 15    | 24    | 20    | 21    | 24    | 24    | 11    | 3     | 1     | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 8.5   | 13.2  | 19.8  | 24.8  | 26.1  | 27.8  | 31.7  | 30.3  | 25.6  | 20.8  | 15.0  | 8.7   | (19) |  |
|                                                                        |             | MCWB   | -0.9  | 1.8   | 5.5   | 8.5   | 10.5  | 14.2  | 16.7  | 16.1  | 14.0  | 9.3   | 4.2   | -0.4  | (20) |  |
|                                                                        | 2%          | DB     | 5.7   | 10.0  | 16.1  | 21.4  | 23.9  | 25.7  | 28.5  | 27.5  | 22.7  | 18.0  | 12.5  | 6.1   | (21) |  |
|                                                                        |             | MCWB   | -2.2  | 0.3   | 3.9   | 7.6   | 10.3  | 13.1  | 15.8  | 15.8  | 12.9  | 8.3   | 2.8   | -1.6  | (22) |  |
|                                                                        | 5%          | DB     | 3.4   | 7.5   | 13.5  | 19.1  | 22.0  | 24.0  | 26.7  | 25.5  | 20.8  | 15.8  | 10.4  | 4.3   | (23) |  |
|                                                                        |             | MCWB   | -3.5  | -0.9  | 2.6   | 6.7   | 9.6   | 12.6  | 15.2  | 15.0  | 12.3  | 7.2   | 2.0   | -2.5  | (24) |  |
|                                                                        | 10%         | DB     | 0.8   | 5.1   | 11.0  | 16.9  | 20.1  | 22.4  | 24.8  | 23.7  | 18.9  | 13.8  | 8.1   | 2.1   | (25) |  |
|                                                                        |             | MCWB   | -4.9  | -2.0  | 1.8   | 5.8   | 9.1   | 12.1  | 14.8  | 14.5  | 11.5  | 6.5   | 1.0   | -3.7  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | -0.8  | 2.2   | 6.2   | 10.2  | 12.7  | 16.5  | 18.6  | 18.4  | 16.3  | 10.7  | 4.8   | -0.1  | (27) |  |
|                                                                        |             | MCDB   | 8.0   | 12.4  | 17.9  | 19.8  | 21.4  | 23.9  | 27.7  | 25.7  | 20.8  | 18.0  | 13.2  | 7.9   | (28) |  |
|                                                                        | 2%          | WB     | -2.2  | 0.5   | 4.4   | 8.6   | 11.5  | 15.0  | 17.4  | 17.2  | 14.7  | 9.2   | 3.5   | -1.4  | (29) |  |
|                                                                        |             | MCDB   | 5.6   | 9.2   | 14.0  | 18.6  | 20.1  | 21.8  | 25.3  | 23.7  | 20.1  | 15.2  | 11.0  | 5.7   | (30) |  |
|                                                                        | 5%          | WB     | -3.5  | -0.7  | 3.2   | 7.6   | 10.7  | 14.1  | 16.5  | 16.4  | 13.4  | 8.2   | 2.4   | -2.5  | (31) |  |
|                                                                        |             | MCDB   | 3.2   | 7.0   | 11.8  | 17.1  | 19.0  | 20.0  | 23.4  | 22.5  | 18.3  | 13.8  | 9.3   | 3.8   | (32) |  |
|                                                                        | 10%         | WB     | -4.8  | -1.8  | 2.2   | 6.6   | 9.9   | 13.3  | 15.6  | 15.6  | 12.4  | 7.2   | 1.3   | -3.6  | (33) |  |
|                                                                        |             | MCDB   | 0.6   | 4.7   | 10.0  | 15.3  | 18.2  | 19.4  | 22.0  | 21.4  | 17.3  | 12.5  | 7.6   | 1.9   | (34) |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 15.6  | 15.3  | 14.3  | 13.5  | 12.8  | 12.3  | 11.8  | 11.5  | 10.8  | 12.7  | 14.2  | 15.0  | (35) |  |
|                                                                        | 5% DB       | MCDBR  | 19.8  | 19.8  | 19.7  | 18.1  | 17.1  | 16.0  | 16.3  | 15.5  | 14.9  | 16.8  | 17.9  | 18.4  | (36) |  |
|                                                                        |             | MCWBR  | 12.9  | 11.6  | 9.6   | 7.3   | 6.0   | 5.5   | 5.3   | 5.0   | 5.8   | 7.6   | 9.4   | 11.5  | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 19.3  | 18.4  | 17.2  | 15.2  | 14.1  | 12.2  | 12.3  | 11.5  | 10.9  | 13.8  | 16.3  | 17.5  | (38) |  |
|                                                                        |             | MCWBR  | 12.6  | 10.8  | 8.4   | 6.2   | 5.3   | 4.7   | 4.5   | 4.1   | 4.6   | 6.4   | 8.8   | 11.1  | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.262 | 0.306 | 0.365 | 0.413 | 0.424 | 0.425 | 0.418 | 0.421 | 0.387 | 0.333 | 0.274 | 0.249 | (40) |  |
|                                                                        | taud        |        | 2.436 | 2.296 | 2.113 | 2.017 | 2.066 | 2.140 | 2.233 | 2.205 | 2.259 | 2.353 | 2.461 | 2.494 | (41) |  |
|                                                                        | Ebn at Noon |        | 944   | 928   | 894   | 868   | 861   | 858   | 861   | 851   | 863   | 885   | 921   | 942   | (42) |  |
|                                                                        | Edh at Noon |        | 84    | 110   | 146   | 171   | 165   | 153   | 139   | 140   | 125   | 103   | 81    | 74    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 3.12  | 3.94  | 4.95  | 5.68  | 5.86  | 5.77  | 5.59  | 5.14  | 4.15  | 3.88  | 3.29  | 2.89  | (44) |  |
|                                                                        | RadStd      |        | 0.18  | 0.17  | 0.28  | 0.29  | 0.33  | 0.30  | 0.52  | 0.35  | 0.32  | 0.32  | 0.20  | 0.11  | (45) |  |

## Historical Trends

|                             |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                         | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| (46) Station Only           | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (5 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page